

Abdul Razak Bilal

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SUMMARY

AI & ML engineering student with hands-on experience in data science, generative AI, and multi-agent systems. Built and deployed end-to-end ML pipelines, RAG-based chatbots, and real-time analytics dashboards using Python, Scikit-learn, IBM WatsonX, AWS, and Streamlit. Completed a remote research internship at the University of North Texas (UNT), contributing to an AI-powered wearable health monitoring system for real-time caregiver decision support. Seeking Data Science or AI Development roles to apply skills in machine learning, predictive modeling, and intelligent system design to real-world problems.

EDUCATION

G. Pulla Reddy Engineering College <i>B.Tech in Computer Science & Engineering (AI & ML)</i>	Kurnool, India CGPA: 8.47 Aug 2024 – Jun 2027
Brindavan Institute of Technology and Science <i>Diploma in Computer Engineering</i>	Kurnool, India Percentage: 95.78% Sep 2021 – May 2024

EXPERIENCE

Remote Research Intern – Data Science & Health Informatics <i>University of North Texas (UNT), USA GPREC-UNT MoU Collaborative Program</i>	May 2026 – Jun 2026 <i>Remote</i>
– Conducted research under Dr. Pretom Roy Ovi (Assistant Professor of Data Science, UNT) through a competitive remote internship under the GPREC-UNT MoU collaborative program. – Contributed to Guardian Twin – a privacy-preserving, AI-powered digital twin system for home patient monitoring and caregiver decision support using consumer-grade wearable devices (Samsung Galaxy Watch 5/6, iHealth FLOW AM6). – Scope covered wearable data pipeline design, real-time anomaly detection, fall detection (accelerometer + ML hybrid), 5-level event classification, and caregiver alert interfaces targeting sub-10-second end-to-end latency.	
Data Analytics Intern <i>iTUX Solutions</i>	Dec 2023 – May 2024 <i>Remote</i>
– Built data analysis and visualization dashboards using Microsoft Power BI to deliver actionable business insights. – Authored training notes, presentations, and technical documentation to simplify complex analytical concepts. – Structured and contributed content for the company website, improving clarity and information hierarchy. – Collaborated cross-functionally with teams to translate data findings into strategic recommendations.	

SKILLS

Programming – Python, Java
Data Science & ML – Pandas, NumPy, Scikit-learn, Matplotlib, XGBoost, OpenCV, Computer Vision, Regression, Classification
Generative AI & Cloud – Prompt Engineering, Gemini API, IBM WatsonX, RAG, Multi-Agent Systems, Groq API, FastAPI, Next.js, AWS (SageMaker, Bedrock, PartyRock), Foundation Models, AI Governance
Frameworks & Tools – Streamlit, Supabase, Power BI, Jupyter Notebook, VS Code, GitHub
Databases – MySQL, SQLite
Soft Skills – Team Leadership, Cross-functional Collaboration, Time Management, Problem-solving, Technical Documentation

PROJECTS

WorldSim 2026 – Live Bayesian FIFA World Cup Prediction Engine [GitHub] [Live]	Jun 2026
– Built and deployed a live FIFA World Cup 2026 tournament simulation engine using a custom Elo rating system with Poisson-based goal modeling, calibrated iteratively against live match results. – Ran 10,000 Monte Carlo simulations per update to compute dynamic championship probabilities across all 32 teams throughout the live tournament. – Deployed a 9-page interactive Streamlit dashboard with real-time standings, match predictions, and group-stage progression tracking.	
Bayyinah Dream Big – Quranic Arabic Learning PWA [Live]	May 2026
– Built a free Duolingo-style Progressive Web App for Quranic Arabic learning as a fan tribute to Nouman Ali Khan's Dream Big intensive. – Engineered 400+ exercises, a full gamification system (XP, streaks, hearts, 45 badges), Arabic TTS, certificate generation, and a global leaderboard.	

- Implemented Google OAuth and Magic Link authentication, PWA offline support, and an admin dashboard using React, TypeScript, Tailwind, Vite, and Supabase.
- MedIQ – Medical Image Quality & Metadata Intelligence System** [GitHub] Feb 2026
- Engineered an automated Quality Control pipeline for medical imaging using Computer Vision techniques (Laplacian Variance, SNR analysis).
 - Analyzed NIH Chest X-ray dataset to correlate image artifacts (blur, noise) with patient metadata for systemic error detection.
 - Deployed a real-time Streamlit dashboard providing instant Accept/Reject diagnostics and disease-pattern visualization.
- WASP – Win Probability Model for IPL** [GitHub] Nov 2025
- Developed an end-to-end win probability model trained on 130k+ IPL ball-by-ball records using Logistic Regression and XGBoost classification models.
 - Engineered match-context features and visualized real-time win probability curves identifying momentum flip moments across innings.
 - Deployed a Streamlit application enabling real-time match prediction and interactive visualization.
- that32bloke – Sports Analytics Blog Platform** [Live] Nov 2025 – Present
- Built a full-featured sports analytics blog platform with CMS, user authentication, and newsletter subscription focused on cricket analytics and data-driven sports storytelling.
 - Conducted security audit identifying and resolving Supabase RLS and XSS vulnerabilities; configured Google Search Console and submitted sitemap for SEO indexing.
- Ryze Pocket Cashier – Point of Sale System** [Live] Jan 2026
- Built a fully functional multi-shop POS system in a single day using Lovable and Supabase with dark mode UI.
 - Included real-time inventory management, billing, and multi-shop switching designed for small business use.
- Play11 Prism – Cricket Performance Analysis Platform** [GitHub] | [Live] May 2025
- Developed an AI-powered platform to optimize T20 playing XI selection based on player performance metrics.
 - Converted a Jupyter Notebook prototype into a production-ready web application with automated best XI generation logic.
- Applied Data Science Lab Projects – WorldQuant University** 2025 – 2026
- Regression & Forecasting: Built models for housing price prediction (Mexico, Buenos Aires), PM2.5 air quality forecasting (Nairobi), and asset volatility modeling using GARCH (India).
 - Classification & Imbalanced Learning: Classified earthquake building damage (Nepal) and predicted corporate bankruptcy (Poland) using ensemble and resampling techniques.
 - Clustering & Experimentation: Segmented US households via K-Means and PCA; designed A/B tests evaluated with chi-square statistical tests.

HACKATHONS

- IBM Dev Day – AI Demystified Hackathon** Feb 2026
- Served as Team Lead and AI Architect to design and deploy a fully functional Multi-Agent Patent Strategy System using IBM WatsonX Orchestrate.
 - Engineered reasoning logic for 5 interacting agents coordinating tasks across patent retrieval, analysis, and strategic recommendation.
 - Collaborated with a globally distributed team, managing architecture decisions and cross-timezone coordination throughout the hackathon.
- AIIgnite 2.0 – Inter-College AI Hackathon** Jan 2026
- Built **MetroBot**, an AI-powered metro query chatbot using Gemini API and Retrieval-Augmented Generation (RAG) to handle real-time passenger queries.
 - Designed document retrieval pipelines grounded on metro data to deliver accurate, context-aware conversational responses.

CERTIFICATIONS

- Machine Learning Foundation – Infosys Springboard** [View Certificate] Jan 2026
- Applied Data Science Lab – WorldQuant University** [View Credly Badge] Jan 2026
- Meta Data Analyst Professional Certificate – Coursera** [View Certificate] Feb 2025
- Google AI Essentials – Coursera** [View Certificate] Aug 2024
- Python Programming – HCL GUVI** [View Certificate]

ACHIEVEMENTS & EXTRACURRICULAR

- Solved 350+ coding problems across competitive programming platforms.
- Recognized as a meritorious student throughout Diploma with 95.78% aggregate.
- Designed and deployed 4+ production web applications end-to-end using HTML, CSS, and JavaScript, hosted on GitHub Pages with polished UI and animations.
- Cricket and Football enthusiast with strong interest in sports analytics; read *Hitting Against the Spin* by Ben Jones & Nathan Leamon.
- AI Art Creator – Admin of ARB AI Arts (@arb.ai.arts, Instagram).